

Quiz (Carolina Reaper)

Q1. Solve the system: $x + y = 3$, $x^2 + y^2 = 9$

- (A) $(x, y) = (1, 2)$
 (B) $(x, y) = (2, 1)$
 (C) $x = 1, y = pm2$
 (D) $x = pm1, y = 2$
 (E) No solution

and $y = x$

- (A) $pm\sqrt{2}$
 (B) $pm1$
 (C) $pm2$
 (D) 0
 (E) 4

Q2. Find the intersection point of $y = x^2 - 4x + 3$ and $y = 2x - 3$

- (A) $(1, -1)$
 (B) $(3, 3)$
 (C) $(2, -1)$
 (D) $(0, 3)$
 (E) $(4, 5)$

Q6. Solve the system: $y = x^2 - 1$, $y = 1 - x$

- (A) $x = 0, y = -1$
 (B) $x = 1, y = 0$
 (C) $x = -1, y = 0$
 (D) $x = 2, y = 3$
 (E) No real solution

Q3. Solve the system: $y = x^2$, $y = 2x - 1$

- (A) $x = 1, y = 1$
 (B) $x = 2, y = 4$
 (C) $x = -1, y = 1$
 (D) $x = 0, y = 0$
 (E) No real solution

Q7. What are the coordinates of the intersection points between $y = x^2 - 2x - 3$ and $y = 1 - x$?

- (A) $(-1, 2)$, $(3, -2)$
 (B) $(-1, -2)$, $(3, 2)$
 (C) $(-2, 3)$, $(1, 0)$
 (D) $(-1, 0)$, $(3, -2)$
 (E) $(0, -3)$, $(2, -1)$

Q4. What is the x -coordinate of the intersection point between $y = x^2 + 2x + 1$ and $y = x + 1$?

- (A) -1
 (B) 0
 (C) 1
 (D) 2
 (E) 3

Q8. Find the value of x for which $x^2 + y^2 = 25$ and $y = 2x - 5$ intersect

- (A) 2
 (B) 4
 (C) -2
 (D) 5
 (E) 0

Q5. Find the y -coordinate of the intersection point of $x^2 + y^2 = 4$ **Open Ended Questions (FRQ)****Q9.** Solve the system: $y = x^2 + 1$, $x + y = 2$. Show all steps and box your final answer.**Q10.** Find the intersection points of $y = x^2 - 4x + 3$ and $y = x - 1$. Explain your reasoning and provide your answer in the required format.

Chef's Tips

- Tip 1: Ensure students show all work and explain reasoning for FRQs
- Tip 2: Remind students to check their solutions in the original equations
- Tip 3: Encourage the use of graphical and algebraic methods to solve systems